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JEE MAIN PHYSICS DPP: THIRD EQUATION OF MOTION

Topic: $v^2 = u^2 + 2as$ & Advanced Problems on Equations of Motion | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A vehicle moving at speed u is brought to rest under uniform braking acceleration in distance s . If the initial velocity is increased to $3u$, the stopping distance under the same deceleration becomes:
- $3s$
 - $9s$
 - $6s$
 - $12s$
- Q2.** A train moving with constant acceleration passes a signal post. Its front engine passes with speed u and the rear guard's van passes with speed v . The speed with which the middle point of the train passes the post is:
- $\frac{u+v}{2}$
 - $\sqrt{uv}$
 - $\sqrt{\frac{u^2+v^2}{2}}$
 - $\frac{2uv}{u+v}$
- Q3.** A bullet fired into a wooden block loses $\frac{1}{3}$ rd of its velocity after penetrating a distance of 5 cm. Assuming constant retarding force, the additional distance it penetrates before coming to rest is:
- 2.5 cm
 - 3.0 cm
 - 5.0 cm
 - 4.0 cm
- Q4.** A stone is dropped from the top of a tower of height h . After falling through distance x , its speed is half of the speed with which it strikes the ground. The value of x is:
- $\frac{h}{4}$
 - $\frac{h}{2}$
 - $\frac{h}{\sqrt{2}}$
 - $\frac{3h}{4}$
- Q5.** A particle moves along the x -axis such that its velocity is given by $v = \sqrt{16 - 8x}$ m/s. The acceleration of the particle is:
- $+4 \text{ m/s}^2$
 - -4 m/s^2
 - -8 m/s^2
 - $+8 \text{ m/s}^2$
- Q6.** An object accelerating uniformly from rest acquires a velocity v over distance s . The velocity acquired over the next distance s is:
- $2v$
 - $\sqrt{3}v$
 - $\sqrt{2}v$
 - $(\sqrt{2} - 1)v$
- Q7.** A particle starts from rest with uniform acceleration. If the velocity is v_1 after distance s_1 and v_2 after distance s_2 , the ratio v_1/v_2 is:
- $\frac{s_1}{s_2}$
 - $\left(\frac{s_1}{s_2}\right)^2$
 - $\frac{s_2}{s_1}$
 - $\sqrt{\frac{s_1}{s_2}}$
- Q8.** A particle moving along a line with constant deceleration travels 12 m while its velocity decreases from 10 m/s to 2 m/s. How much further will it travel before coming to rest?
- 0.5 m
 - 1.0 m
 - 1.5 m
 - 2.0 m
- Q9.** A body is thrown vertically upward with velocity u . The velocity of the body at half the maximum height is:
- $\frac{u}{2}$
 - $\frac{u}{\sqrt{2}}$
 - $\frac{\sqrt{3}u}{2}$
 - $\frac{u}{4}$
- Q10.** The reaction time of an automobile driver is 0.6 s. If the car can decelerate at 6 m/s^2 and moves at 54 km/h, the total distance covered from the instant the danger is spotted until the car comes to rest is:
- 18.75 m
 - 24.25 m
 - 27.75 m

(d) 32.50 m

Q11. Two cars P and Q start from a point at the same time in a straight line. P moves with uniform acceleration a_1 while Q moves with uniform acceleration a_2 ($a_1 > a_2$). When car P has covered distance S , the speed of P relative to Q is:

- (a) $\sqrt{2(a_1 - a_2)S}$
- (b) $\sqrt{2a_1S} - \sqrt{2a_2S}$
- (c) $\sqrt{2(a_1 + a_2)S}$
- (d) $\sqrt{2S}(\sqrt{a_1} - \sqrt{a_2})$

Q12. A lift ascends with constant acceleration a . A passenger drops a coin when the lift is at height H moving upwards with speed V . The speed with which the coin strikes the ground floor of the building is:

- (a) $\sqrt{V^2 + 2gH}$
- (b) $\sqrt{V^2 - 2gH}$
- (c) $\sqrt{2gH}$
- (d) $\sqrt{V^2 + 2(g + a)H}$

Q13. A body accelerates uniformly from rest at 2 m/s^2 for some distance, then immediately decelerates uniformly at 4 m/s^2 to rest. If the total distance travelled is 60 m, the maximum speed attained is:

- (a) $4\sqrt{15} \text{ m/s}$
- (b) $4\sqrt{10} \text{ m/s}$
- (c) 10 m/s
- (d) 20 m/s

Q14. A particle travels through three consecutive displacements s_1, s_2 , and s_3 with uniform acceleration. If the velocity increases by equal amounts in each interval ($v_1 - v_0 = v_2 - v_1 = v_3 - v_2 = \Delta v$), the relation between s_1, s_2 , and s_3 is:

- (a) $s_2 = \frac{s_1 + s_3}{2}$
- (b) $s_2 = \sqrt{s_1 s_3}$
- (c) $s_3 - s_2 = s_2 - s_1$
- (d) $s_1 + s_3 = 2s_2$

Q15. A target consists of two parallel planks of thicknesses x_1 and x_2 . A bullet enters the first plank at speed u , emerges with speed v_1 , and emerges from the second plank with speed v_2 . Assuming equal constant retarding forces, the ratio $\frac{u^2 - v_1^2}{v_1^2 - v_2^2}$ is:

- (a) $\left(\frac{x_1}{x_2}\right)^2$
- (b) $\sqrt{\frac{x_1}{x_2}}$
- (c) $\frac{x_2}{x_1}$
- (d) $\frac{x_1}{x_2}$

Q16. A ball is released from the top of an inclined plane of length L . The ratio of the time taken to travel the first half of the length to that for the second half is:

- (a) $\frac{1}{\sqrt{2} - 1}$
- (b) $\sqrt{2} - 1$
- (c) $\frac{1}{\sqrt{2}}$
- (d) $\sqrt{2} + 1$

Q17. A particle moving with constant acceleration has velocities u and v at points A and B . If C is a point dividing AB such that $AC : CB = 1 : 3$, the speed of the particle at C is:

- (a) $\sqrt{\frac{u^2 + 3v^2}{4}}$
- (b) $\sqrt{\frac{3u^2 + v^2}{4}}$
- (c) $\frac{3u + v}{4}$
- (d) $\sqrt{\frac{u^2 + v^2}{2}}$

Q18. A particle moving along the x -axis has acceleration $a = 2x$. If its velocity at $x = 1 \text{ m}$ is 2 m/s , its velocity at $x = 3 \text{ m}$ is:

- (a) 4 m/s
- (b) $2\sqrt{3} \text{ m/s}$
- (c) $2\sqrt{5} \text{ m/s}$
- (d) 6 m/s

Q19. A particle starts from rest and moves with uniform acceleration a . The displacement during the interval in which its velocity changes from v_1 to v_2 is:

- (a) $\frac{v_2 - v_1}{a}$
- (b) $\frac{v_2^2 - v_1^2}{a}$
- (c) $\frac{(v_2 - v_1)^2}{2a}$
- (d) $\frac{v_2^2 - v_1^2}{2a}$

Q20. A body sliding down a smooth inclined plane of inclination θ attains a speed v at the bottom. If the inclination is increased to 90° (vertical free fall from the same vertical height h), the speed at the bottom will be:

- (a) Equal to v
- (b) $v \sin \theta$
- (c) $\frac{v}{\sin \theta}$
- (d) $v \cos \theta$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

From $v^2 = u^2 - 2as$ with $v = 0$:

Stopping distance $s = \frac{u^2}{2a} \propto u^2$.

When speed is increased to $3u$:

$$s' = \frac{(3u)^2}{2a} = 9 \left(\frac{u^2}{2a} \right) = 9s$$

Sol 2. Correct Option: (c)

Let total train length be L .

$$v^2 = u^2 + 2aL \implies 2aL = v^2 - u^2.$$

For the midpoint ($s = L/2$):

$$v_{\text{mid}}^2 = u^2 + 2a \left(\frac{L}{2} \right) = u^2 + \frac{2aL}{2}$$

$$v_{\text{mid}}^2 = u^2 + \frac{v^2 - u^2}{2} = \frac{u^2 + v^2}{2}$$

$$v_{\text{mid}} = \sqrt{\frac{u^2 + v^2}{2}}$$

Sol 3. Correct Option: (d)

Initial speed u . Speed after $s_1 = 5$ cm is

$$v_1 = u - \frac{u}{3} = \frac{2u}{3}.$$

$$v_1^2 = u^2 - 2a(5) \implies \left(\frac{2u}{3} \right)^2 = u^2 - 10a$$

$$10a = u^2 - \frac{4}{9}u^2 = \frac{5}{9}u^2 \implies 2a = \frac{u^2}{9}$$

For additional distance s_2 to come to rest ($v = 0$):

$$0 = v_1^2 - 2as_2 \implies \left(\frac{2u}{3} \right)^2 = 2as_2$$

$$\frac{4}{9}u^2 = \left(\frac{u^2}{9} \right) s_2 \implies s_2 = 4.0 \text{ cm}$$

Sol 4. Correct Option: (a)

Final ground speed: $v_g^2 = 0 + 2gh = 2gh$.

Speed at distance x : $v_x^2 = 2gx$.

$$\text{Given } v_x = \frac{1}{2}v_g \implies v_x^2 = \frac{1}{4}v_g^2:$$

$$2gx = \frac{1}{4}(2gh) \implies x = \frac{h}{4}$$

Sol 5. Correct Option: (b)

$$v^2 = 16 - 8x.$$

Differentiating with respect to x :

$$2v \frac{dv}{dx} = -8 \implies a = v \frac{dv}{dx} = -4 \text{ m/s}^2$$

Sol 6. Correct Option: (c)

First segment ($u = 0$ to v over s):

$$v^2 = 0 + 2as \implies 2as = v^2$$

Total motion over distance $2s$ with final speed v' :

$$(v')^2 = 0 + 2a(2s) = 2(2as) = 2v^2 \implies v' = \sqrt{2}v$$

Sol 7. Correct Option: (d)

Using $v^2 = 2as$ ($u = 0$):

$$v_1 = \sqrt{2as_1} \text{ and } v_2 = \sqrt{2as_2}.$$

Ratio:

$$\frac{v_1}{v_2} = \sqrt{\frac{s_1}{s_2}}$$

Sol 8. Correct Option: (a)

From $v^2 = u^2 - 2as$:

$$2^2 = 10^2 - 2a(12) \implies 4 = 100 - 24a \implies 24a = 96 \implies 2a = 8.$$

For remaining distance s' from $v = 2$ m/s to rest ($v = 0$):

$$0 = 2^2 - 2as' \implies 4 = 8s' \implies s' = 0.5 \text{ m}$$

Sol 9. Correct Option: (b)

Maximum height: $H = \frac{u^2}{2g}$.

At height $h = H/2 = \frac{u^2}{4g}$:

$$v^2 = u^2 - 2g \left(\frac{u^2}{4g} \right) = u^2 - \frac{u^2}{2} = \frac{u^2}{2} \implies v = \frac{u}{\sqrt{2}}$$

Sol 10. Correct Option: (c)

Speed $u = 54 \text{ km/h} = 54 \times \frac{5}{18} = 15 \text{ m/s}$.

• Distance during reaction time:

$$s_1 = ut_r = 15 \times 0.6 = 9.0 \text{ m}$$

• Braking distance:

$$s_2 = \frac{u^2}{2a} = \frac{(15)^2}{2(6)} = \frac{225}{12} = 18.75 \text{ m}$$

Total distance $= s_1 + s_2 = 9.0 + 18.75 = 27.75 \text{ m}$.

Sol 11. Correct Option: (d)

For car P : $v_P^2 = 2a_1S \Rightarrow v_P = \sqrt{2a_1S}$.
 Time taken by P to cover S : $S = \frac{1}{2}a_1t^2 \Rightarrow t = \sqrt{\frac{2S}{a_1}}$.
 Speed of car Q at that same time t : $v_Q = \sqrt{2a_2S}$,
 and $v_Q = a_2\sqrt{\frac{2S}{a_1}}$.
 $v_{\text{rel}} = v_P - v_Q = \sqrt{2S}(\sqrt{a_1} - \sqrt{a_2})$.

Sol 12. Correct Option: (a)

Initial velocity of coin relative to ground is $u = +V$ (upwards).
 When it strikes ground at displacement $y = -H$:

$$v^2 = V^2 + 2gH \Rightarrow v = \sqrt{V^2 + 2gH}$$

Sol 13. Correct Option: (b)

Let acceleration distance be s_1 and deceleration be s_2 ($s_1 + s_2 = 60$).

$$v_{\text{max}}^2 = 2(2)s_1 = 4s_1 \Rightarrow s_1 = \frac{v_{\text{max}}^2}{4}$$

$$0 = v_{\text{max}}^2 - 2(4)s_2 \Rightarrow s_2 = \frac{v_{\text{max}}^2}{8}$$

$$s_1 + s_2 = \frac{v_{\text{max}}^2}{4} + \frac{v_{\text{max}}^2}{8} = \frac{3v_{\text{max}}^2}{8} = 60$$

$$v_{\text{max}}^2 = \frac{60 \times 8}{3} = 160$$

$$v_{\text{max}} = \sqrt{160} = 4\sqrt{10} \text{ m/s}$$

Sol 14. Correct Option: (c)

$v_1^2 - v_0^2 = 2as_1$, $v_2^2 - v_1^2 = 2as_2$, $v_3^2 - v_2^2 = 2as_3$.
 Since $v_k = v_0 + k\Delta v$ (AP in velocity):
 $(v_1 - v_0)(v_1 + v_0) = 2as_1 \Rightarrow \Delta v(2v_0 + \Delta v) = 2as_1$
 $\Delta v(2v_0 + 3\Delta v) = 2as_2 \Rightarrow s_2 - s_1 = \frac{2(\Delta v)^2}{2a}$
 $s_3 - s_2 = \frac{2(\Delta v)^2}{2a}$.
 Thus, $s_3 - s_2 = s_2 - s_1$ (displacements form an AP).

Sol 15. Correct Option: (d)

Plank 1: $v_1^2 = u^2 - 2ax_1 \Rightarrow u^2 - v_1^2 = 2ax_1$.
 Plank 2: $v_2^2 = v_1^2 - 2ax_2 \Rightarrow v_1^2 - v_2^2 = 2ax_2$.
 Dividing: $\frac{u^2 - v_1^2}{v_1^2 - v_2^2} = \frac{x_1}{x_2}$.

Sol 16. Correct Option: (a)

Total time for length L : $T = \sqrt{2L/a}$.
 Time for first half ($L/2$): $t_1 = \sqrt{2(L/2)/a} = \frac{T}{\sqrt{2}}$.
 Time for second half:

$$t_2 = T - t_1 = T \left(1 - \frac{1}{\sqrt{2}}\right) = T \left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$\text{Ratio: } \frac{t_1}{t_2} = \frac{T/\sqrt{2}}{T(\sqrt{2}-1)/\sqrt{2}} = \frac{1}{\sqrt{2}-1}$$

Sol 17. Correct Option: (b)

Let total distance $AB = 4s$, so $AC = s$ and

$$CB = 3s.$$

$$v^2 = u^2 + 2a(4s) \Rightarrow 2as = \frac{v^2 - u^2}{4}.$$

At point C ($s = AC$):

$$v_C^2 = u^2 + 2as = \frac{3u^2 + v^2}{4}$$

$$v_C = \sqrt{\frac{3u^2 + v^2}{4}}$$

Sol 18. Correct Option: (c)

$$a = v \frac{dv}{dx} = 2x \Rightarrow v dv = 2x dx.$$

Integrating from $x = 1$ ($v = 2$) to $x = 3$ ($v = v_f$):

$$\left[\frac{v^2}{2}\right]_2^{v_f} = [x^2]_1^3 \Rightarrow \frac{v_f^2 - 4}{2} = 9 - 1 = 8$$

$$v_f^2 - 4 = 16 \Rightarrow v_f = \sqrt{20} = 2\sqrt{5} \text{ m/s}$$

Sol 19. Correct Option: (d)

From $v^2 = u^2 + 2as$, setting $u = v_1$ and $v = v_2$:

$$v_2^2 = v_1^2 + 2as \Rightarrow s = \frac{v_2^2 - v_1^2}{2a}$$

Sol 20. Correct Option: (a)

For an incline of height h and length $L = h/\sin\theta$:
 $a = g \sin\theta$.

$$v^2 = 2aL = 2(g \sin\theta) \left(\frac{h}{\sin\theta}\right) = 2gh \Rightarrow v = \sqrt{2gh}.$$

This speed depends only on the vertical drop height h , independent of θ .

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